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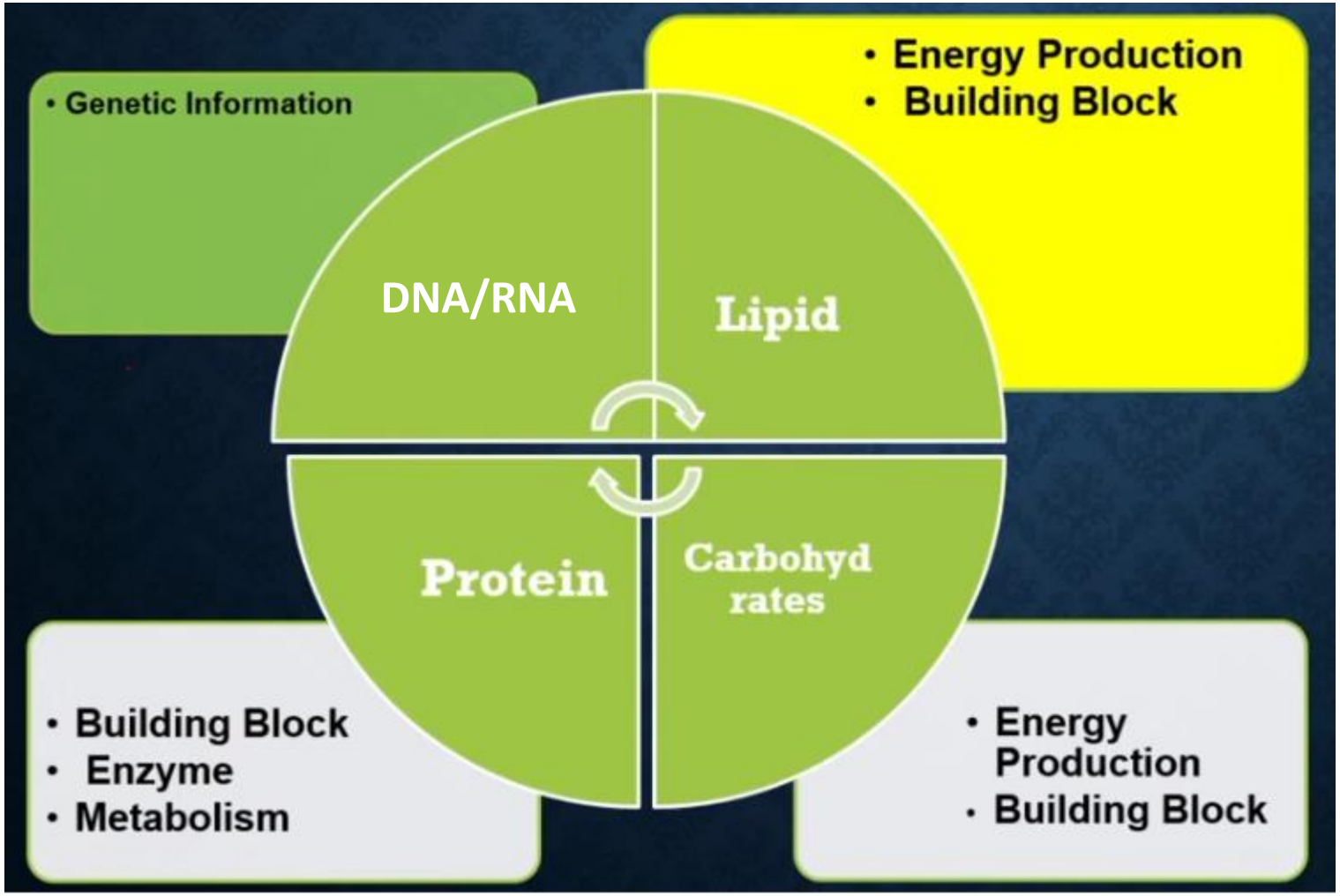
INDIAN INSTITUTE OF TECHNOLOGY DELHI
ABU DHABI

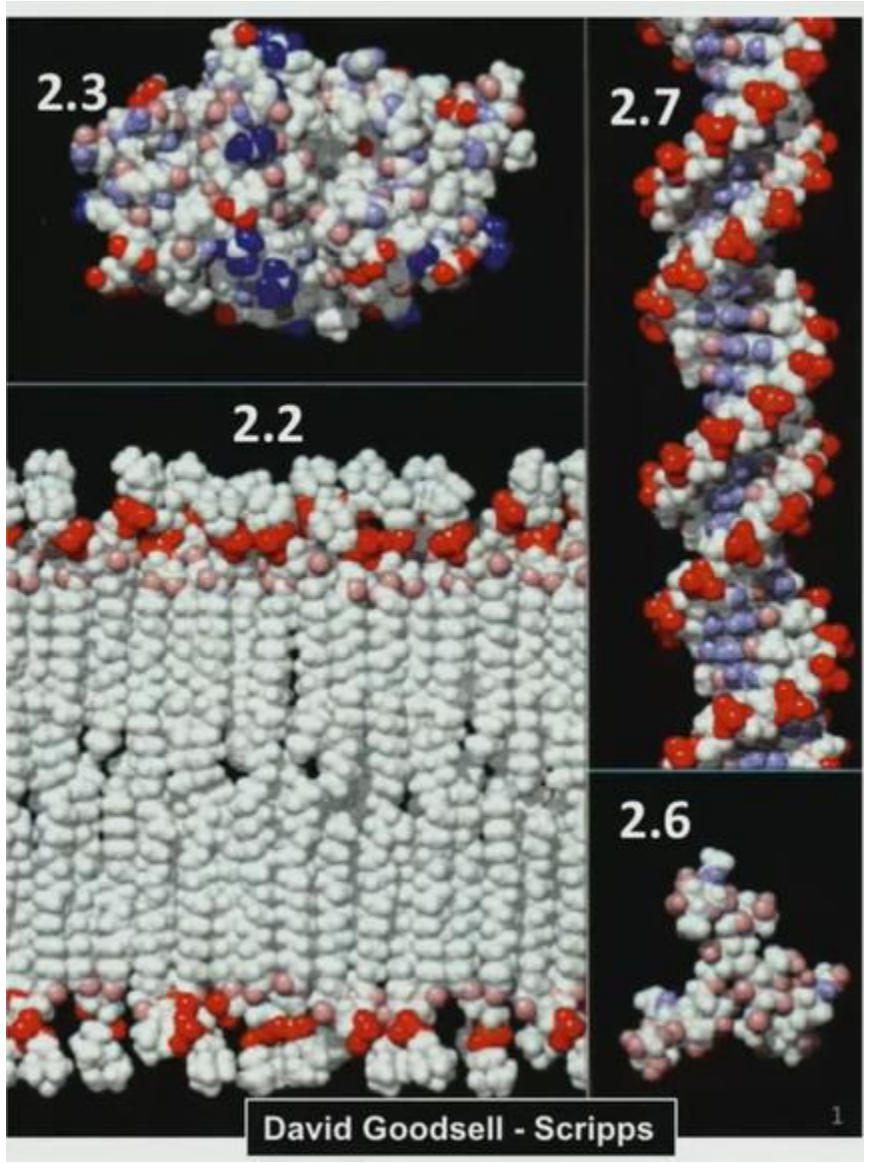
Biomolecules – Nucleic acids

Lecture 5

Prof. Saurabh Raj
Kusuma School of Biological Sciences
IIT Delhi

Molecule of Life

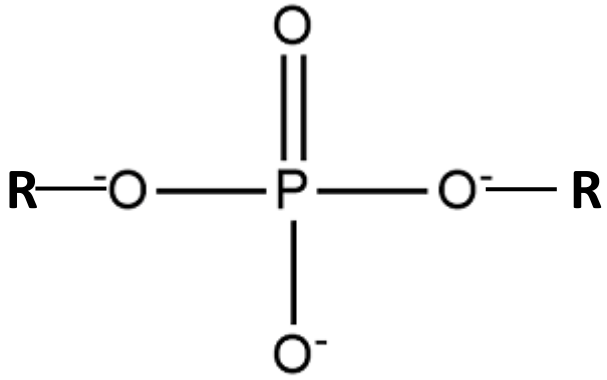




Nucleic acid

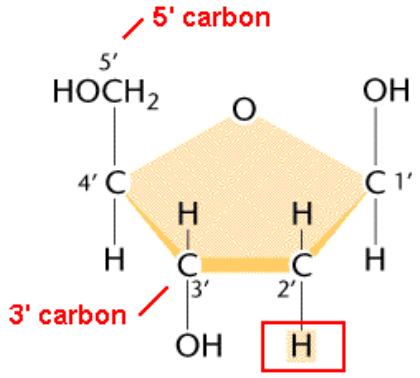
- Most of the organisms has DNA as the genetic material, whereas a minor fraction (Viruses) has RNA as genetic material
- DNA or RNA are biopolymer and are acidic in nature.
- In eukaryotic cells, nucleic acid is present with the nucleus whereas in prokaryotic cells, it is present as free form in the cell.

Nucleotides

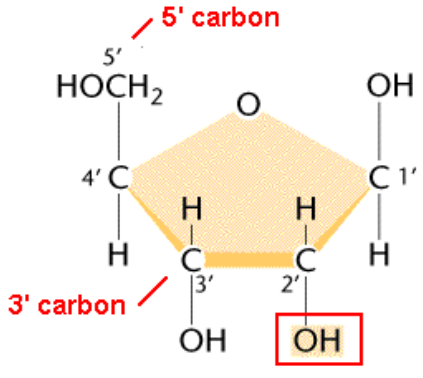


Phosphate

+



2-Deoxyribose



Ribose

+ Nucleobases

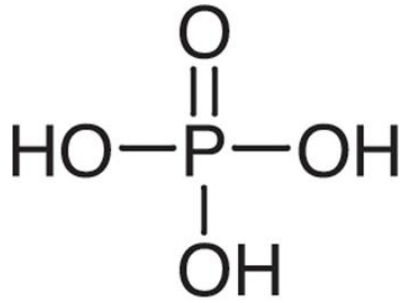
C-Pentose Sugar

(Klug & Cummings 1997)

Nucloesides

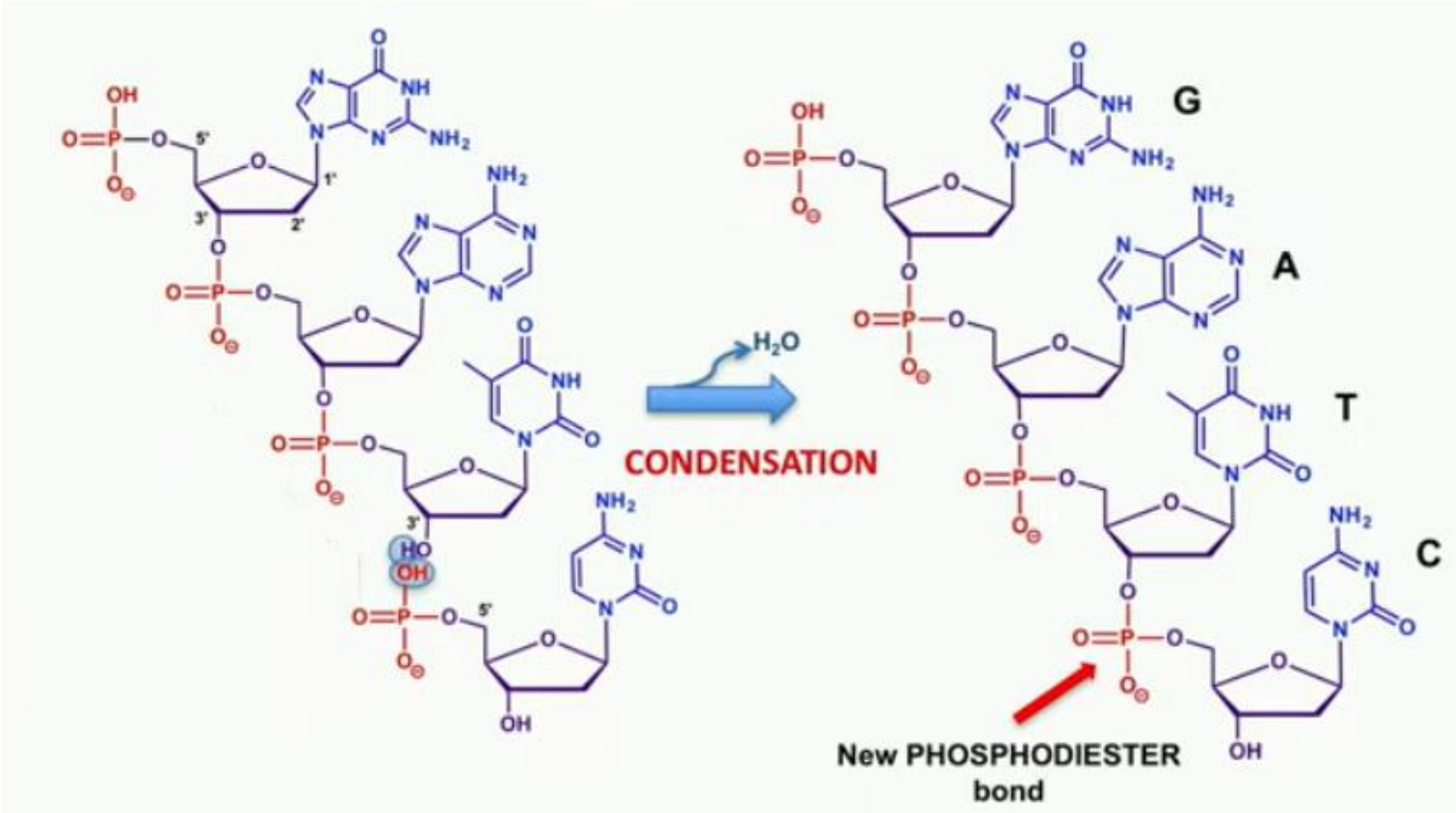
NA is composed of 3 components:

1) Phosphate
2) Nucleobases
3) Sugar



Phosphoric Acid

dsDNA structure



Origin story for dsDNA structure

Erwin Chargaff predicted in 1952 that in all living organisms, the ratio of purines and pyrimidines is 1:1. This is known as Chargaff rule

Chargaff's Data (1950-51)

	Base composition, mole percent				Base ratios			Asymmetry ratio
	A	G	C	T	A/T	G/C	Pu/Py	$\frac{A+T}{G+C}$
Animals								
Man	30.9	19.9	19.8	29.4	1.05	1.00	1.04	1.52
Sheep	29.3	21.4	21.0	28.3	1.03	1.02	1.03	1.36
Hen	28.8	20.5	21.5	29.2	1.02	0.95	0.97	1.38
Turtle	29.7	22.0	21.3	27.9	1.05	1.03	1.00	1.31
Salmon	29.7	20.8	20.4	29.1	1.02	1.02	1.02	1.43
Sea urchin	32.8	17.7	17.3	32.1	1.02	1.02	1.02	1.58
Locust	29.3	20.5	20.7	29.3	1.00	1.00	1.00	1.41
Plants								
Wheat germ	27.3	22.7	22.8	27.1	1.01	1.00	1.00	1.19
Yeast	31.3	18.7	17.1	32.9	0.95	1.09	1.00	1.79
Aspergillus niger (mold)	25.0	25.1	25.0	24.9	1.00	1.00	1.00	1.00
Bacteria								
E. coli	24.7	26.0	25.7	23.6	1.04	1.01	1.03	0.93
Staphylococcus aureus	30.8	21.0	19.0	29.2	1.05	1.11	1.07	1.50
Clostridium perfringens	36.9	14.0	12.8	36.3	1.01	1.09	1.04	2.70
Brucella abortus	21.0	29.0	28.9	21.1	1.00	1.00	1.00	0.72
Sarcina lutea	13.4	37.1	37.1	12.4	1.08	1.00	1.04	0.35
Bacteriophages								
T7	26.0	24.0	24.0	26.0	1.00	1.00	1.00	1.08
λ	21.3	28.6	27.2	22.9	0.92	1.05	1.00	0.79
ϕX174, viral	24.6	24.1	18.5	32.7	0.75	1.30	0.95	1.34
ϕX174, replicative	26.3	22.3	22.3	26.4	1.00	1.00	1.00	1.18

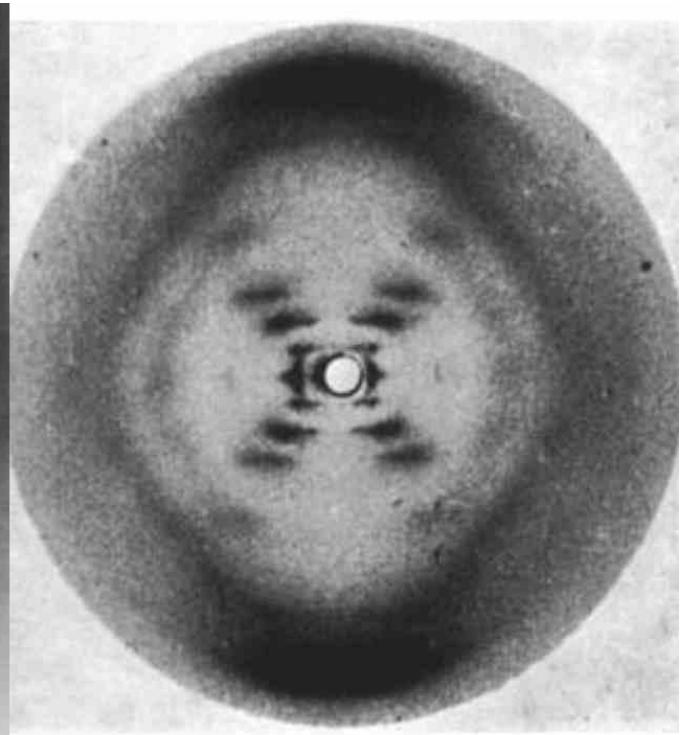
Despite variations in the number of A, T, G and C, for each purine there is a pyrimidine

https://pubs.acs.org/jacsat/article-abstract/71/4/1513/626038/THE-NUCLEOTIDE-COMPOSITION-OF-RIBONUCLEIC-ACIDS?redirectedFrom=fulltext

Origin story for dsDNA structure



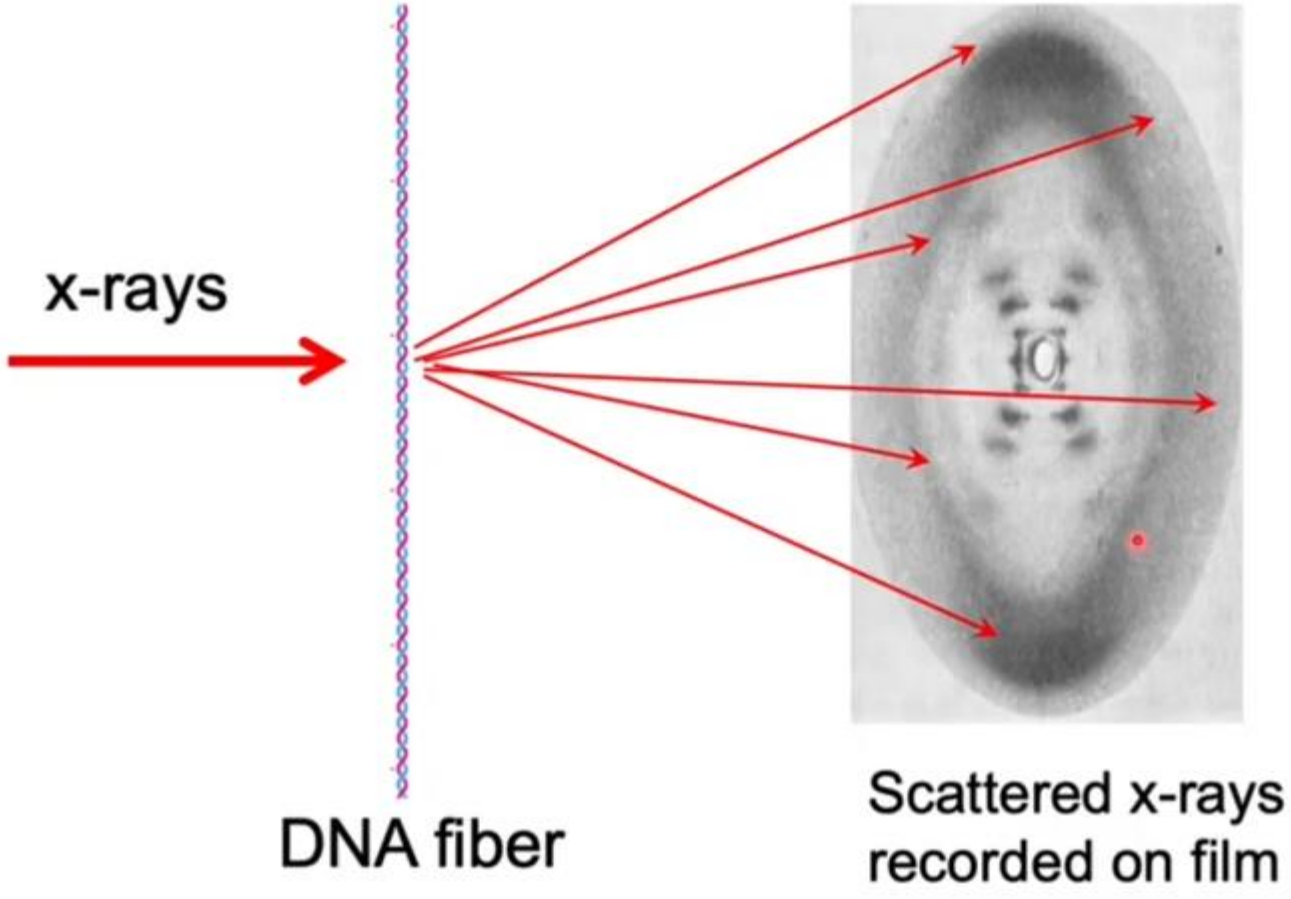
Rosalind Franklin
(1920-1958)



"Photograph 51"

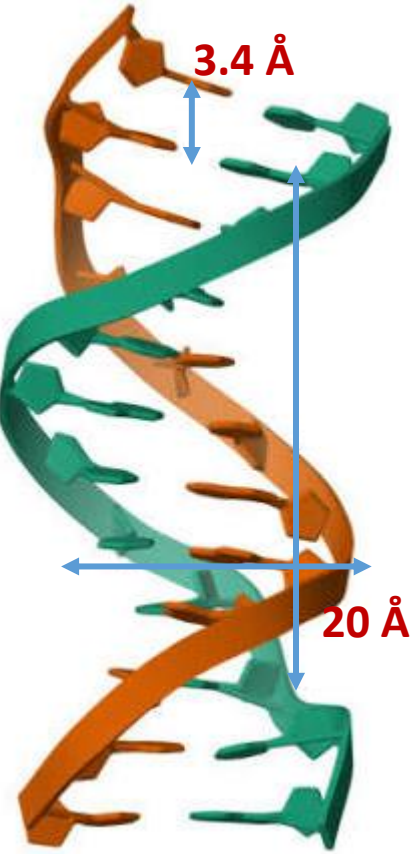
X-Ray diffraction image of crystalline DNA gave a critical dimension to enable model building.

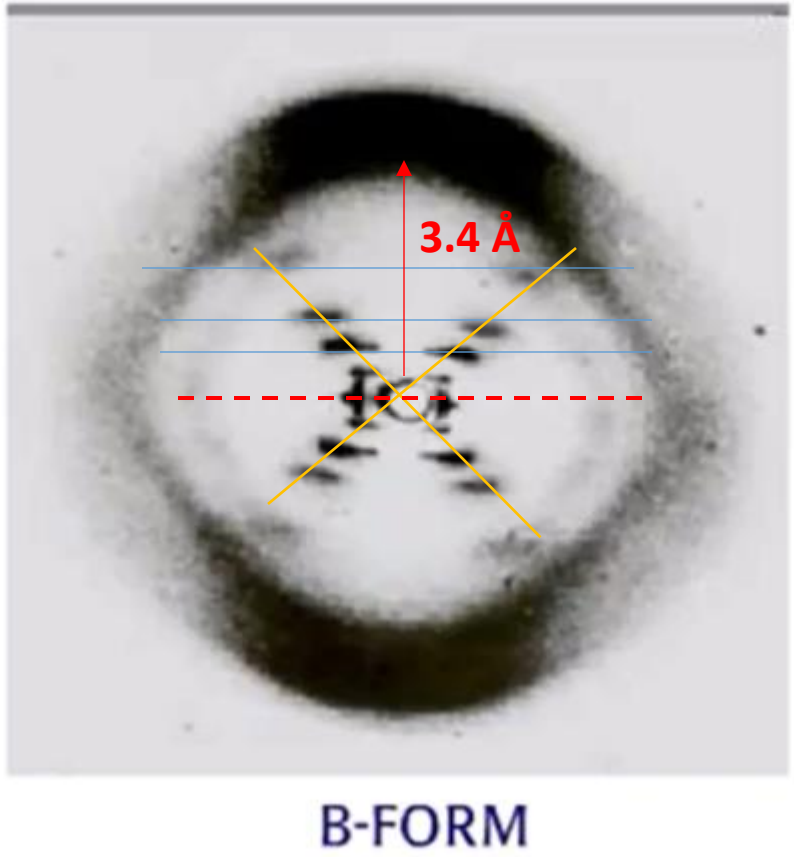
Origin story for dsDNA structure



X-Ray diffraction image of crystalline DNA gave a critical dimension to enable model building.

Origin story for dsDNA structure



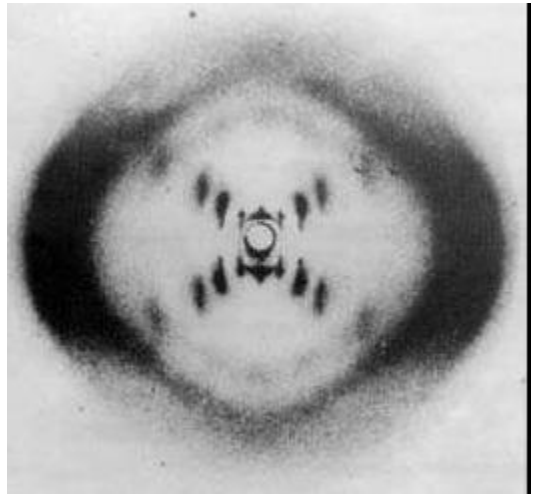


- Regular X-Ray pattern suggested that the dimension of helix is also consistent.
- e.g. the dimension of the helix stay the same
- The horizontal line denotes the helical turn that is 34 Å.
- Distance between middle of the X-Ray pattern to the top corresponds to 3.4 A. This corresponds to the distance between two stacked basepair

https://journals.iucr.org/q/issues/1953/08-09/00/a00979/a00979.pdf
https://collections.nlm.nih.gov/catalog/nlm:nlmuid-101584582X381-doc

Origin story for dsDNA structure

- DNA consists of two chains of nucleotides in a ladder-like structure which is twisted (Double Helix)
- Used data of *M.H.F. Wilkins and Rosalind Franklin*, ***early 50's.***
- Wilkins and Franklin studied the structure of DNA crystals using X-rays.
- The X pattern suggested the structure of DNA was a helix.
- Used data of ***Erwin Chargaff, 1940's and early 50's***
  - Chargaff's Rule: His data showed that in each species, the percent of A equals the percent of T, and the percent of G equals the percent of C

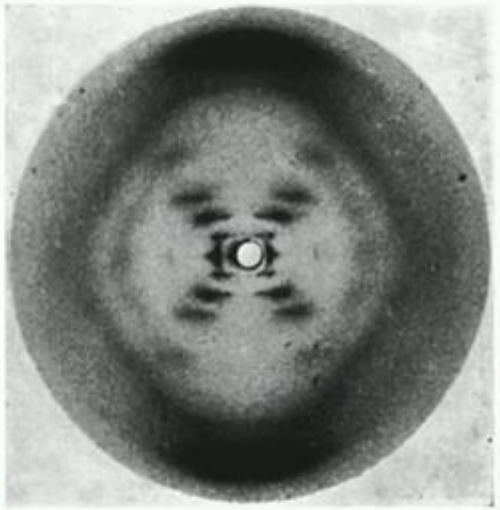


Origin story for dsDNA structure

Rosalind Franklin
(1920-1958)



X-Ray diffraction image of crystalline DNA
gave a critical dimension to enable model building



"Photograph 51"

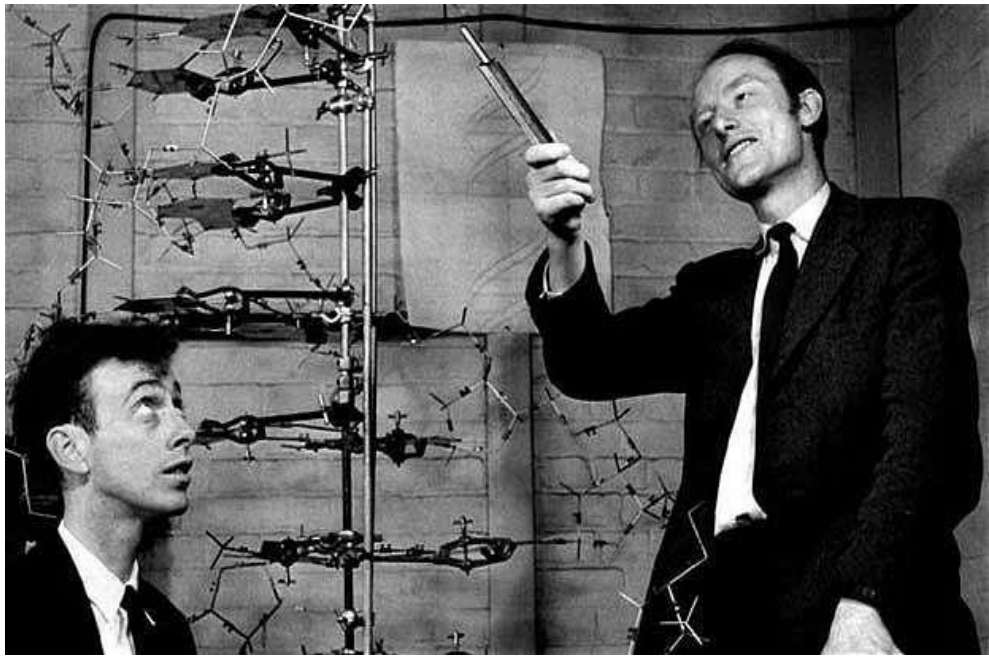


+ Chargaff data (1 PURINE : 1 PYRIMIDINE)

Origin story for dsDNA structure

Watson and Crick's Work

- In 1951 James Watson traveled from the United States to work with Francis Crick at Cambridge University.
- Watson and Crick used the "***Model Building***" approach.
- They physically built models out of wire, sheet metal, nuts and bolts to come up with the structure of DNA.



A. BARRINGTON BROWN, © GONVILLE & CAIUS COLLEGE / SCIENCE PHOTO LIBRARY

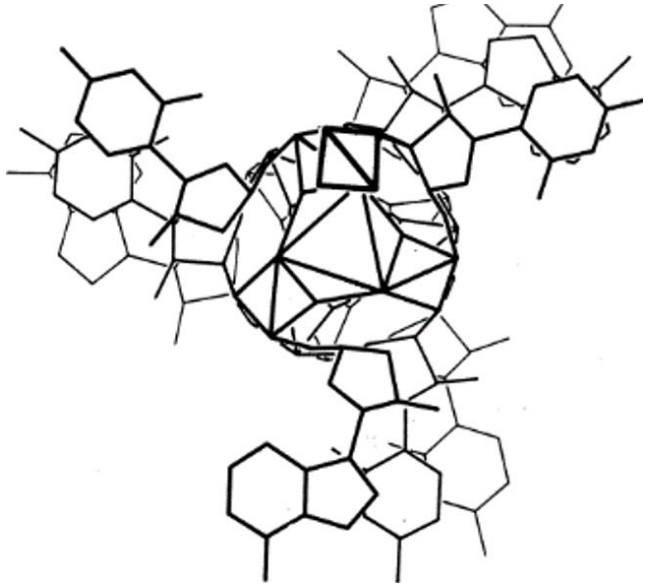
Why did they build models?

"*Sometimes the fingers can grasp what the mind cannot*"
(Biology the Science of Life)

Watson and Crick Model:

- The sides of the ladder are made up of alternating molecules of *phosphate and deoxyribose*.
- The bases make up the rungs of the ladder are attracted by a weak chemical bonds called *hydrogen bonds*.
- The DNA double helix is *anti-parallel*, which means that the 5' end of one strand is paired with the 3' end of its complementary strand (and vice versa).
- Two hydrogen bonds connect T to A; three hydrogen bonds connect G to C.

DNA model proposed by Pauling and Corey:



A PROPOSED STRUCTURE FOR THE NUCLEIC ACIDS

By Linus Pauling and Robert B. Corey

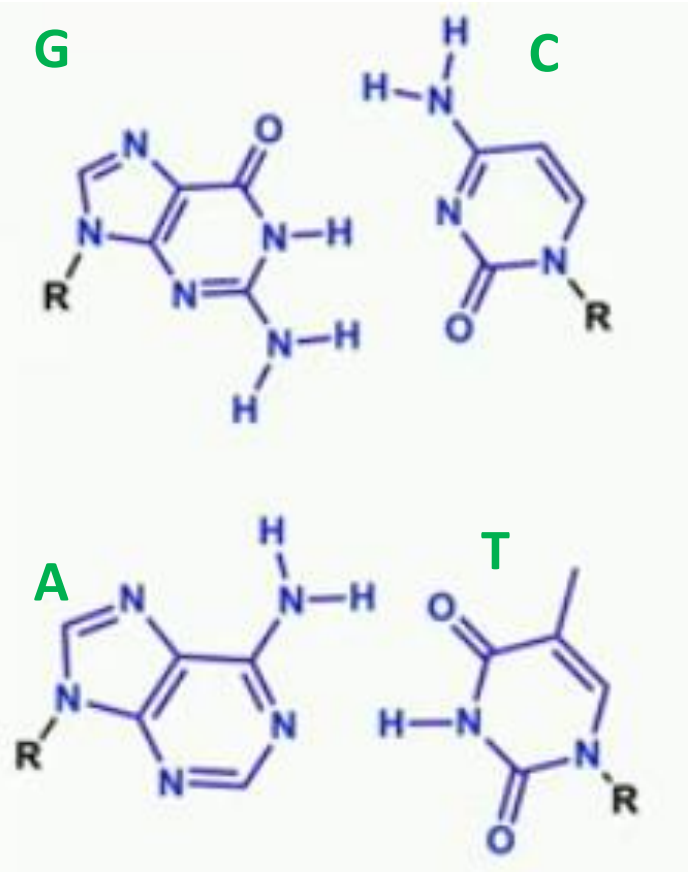
Gates and Crellin Laboratories of Chemistry,* California Institute of Technology

Communicated December 31, 1952

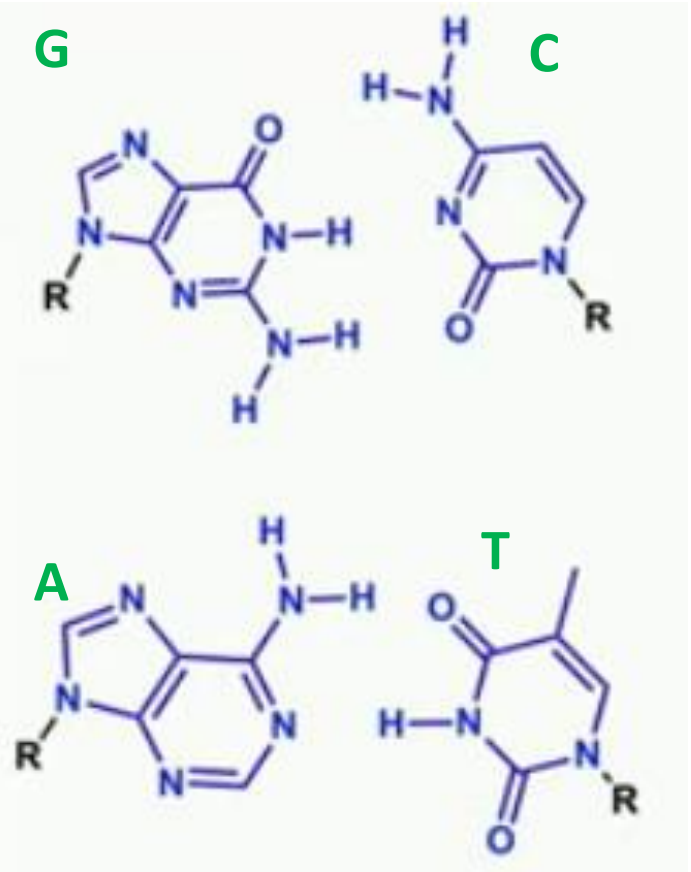
Pauling has predicted in 1948 that gene would consist of two complimentary parts!

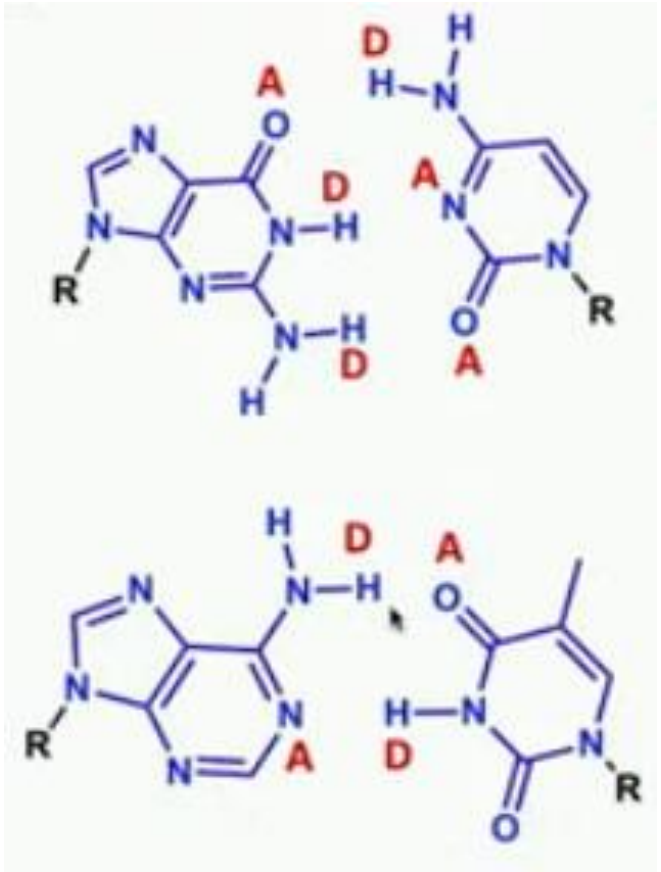
"If the structure that serves as a template (the gene or virus molecule) consists of, say, two parts, which are themselves complementary in structure, then each of these parts can serve as the mould for the production of a replica of the other part, and the complex of two complementary parts thus can serve as the mould for the production of duplicates of itself."

dsDNA structure

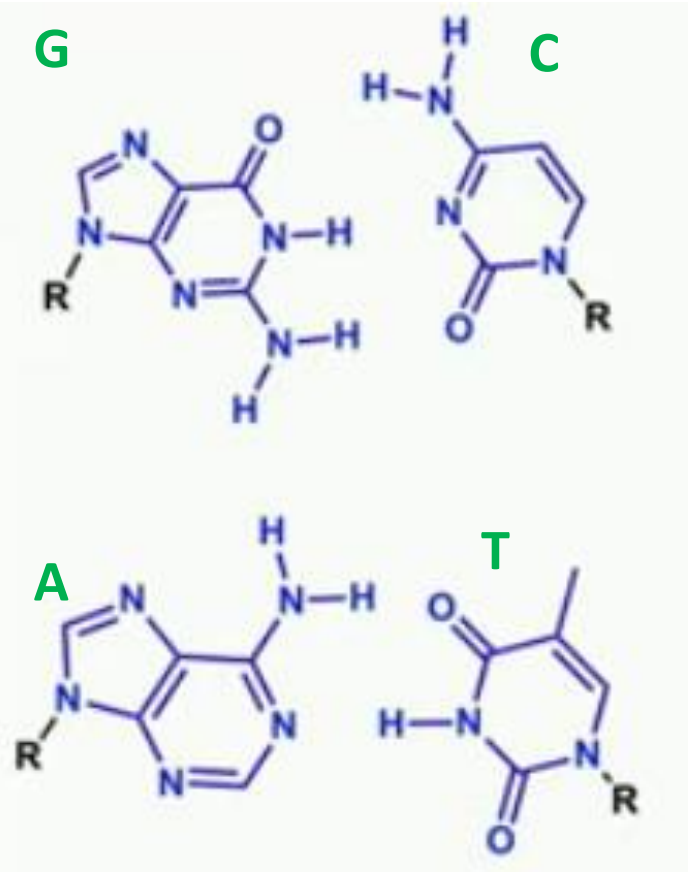


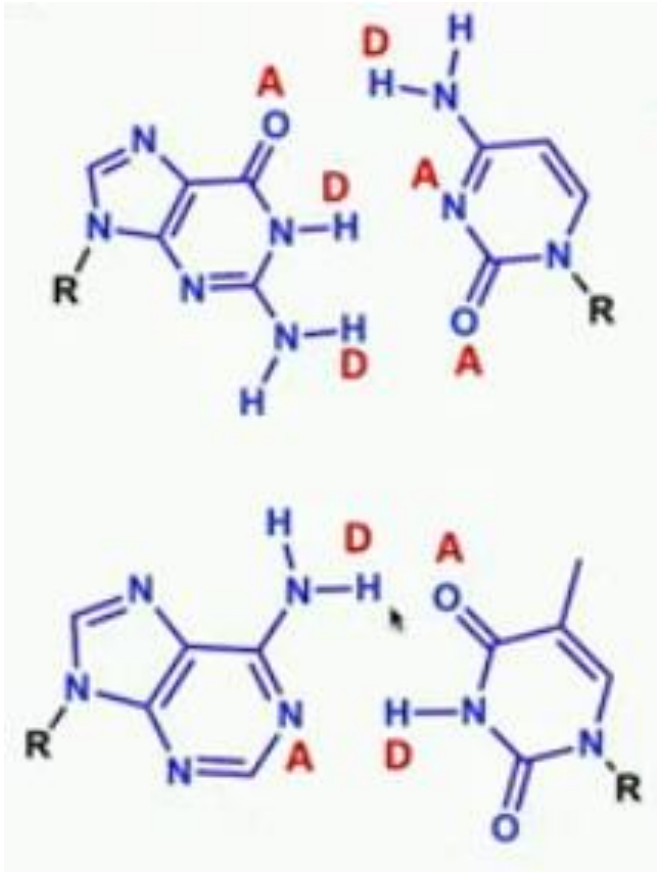
dsDNA structure

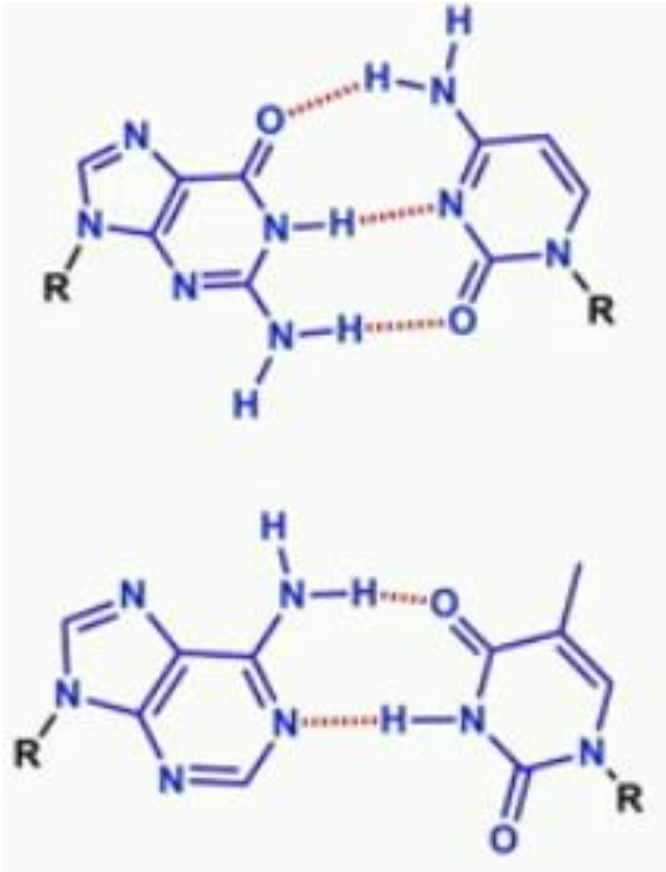




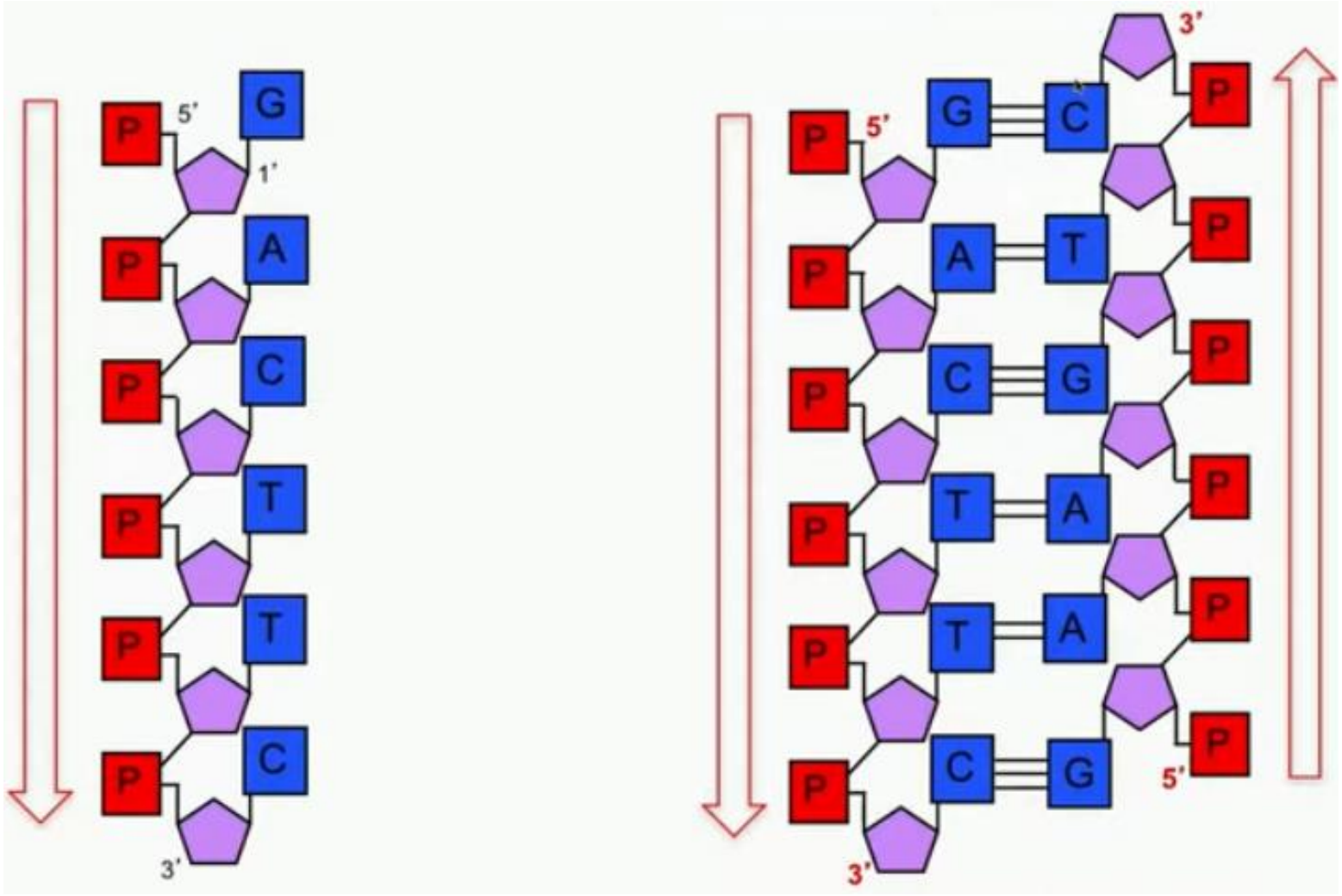
dsDNA structure







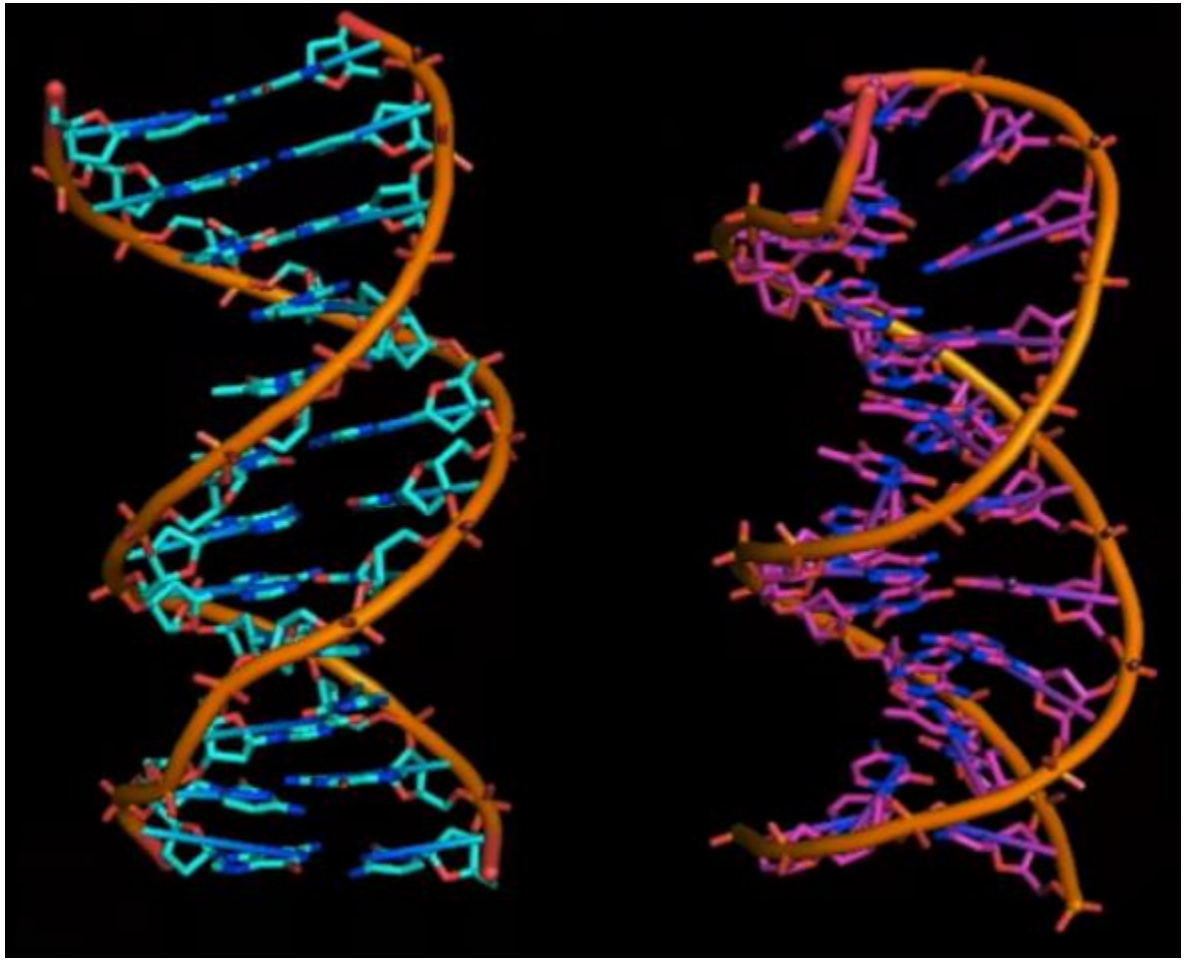
dsDNA structure is antiparallel



5’ pGACTTC 3’

5’ pGACTTC 3’
3’ CTGAAGp 5’

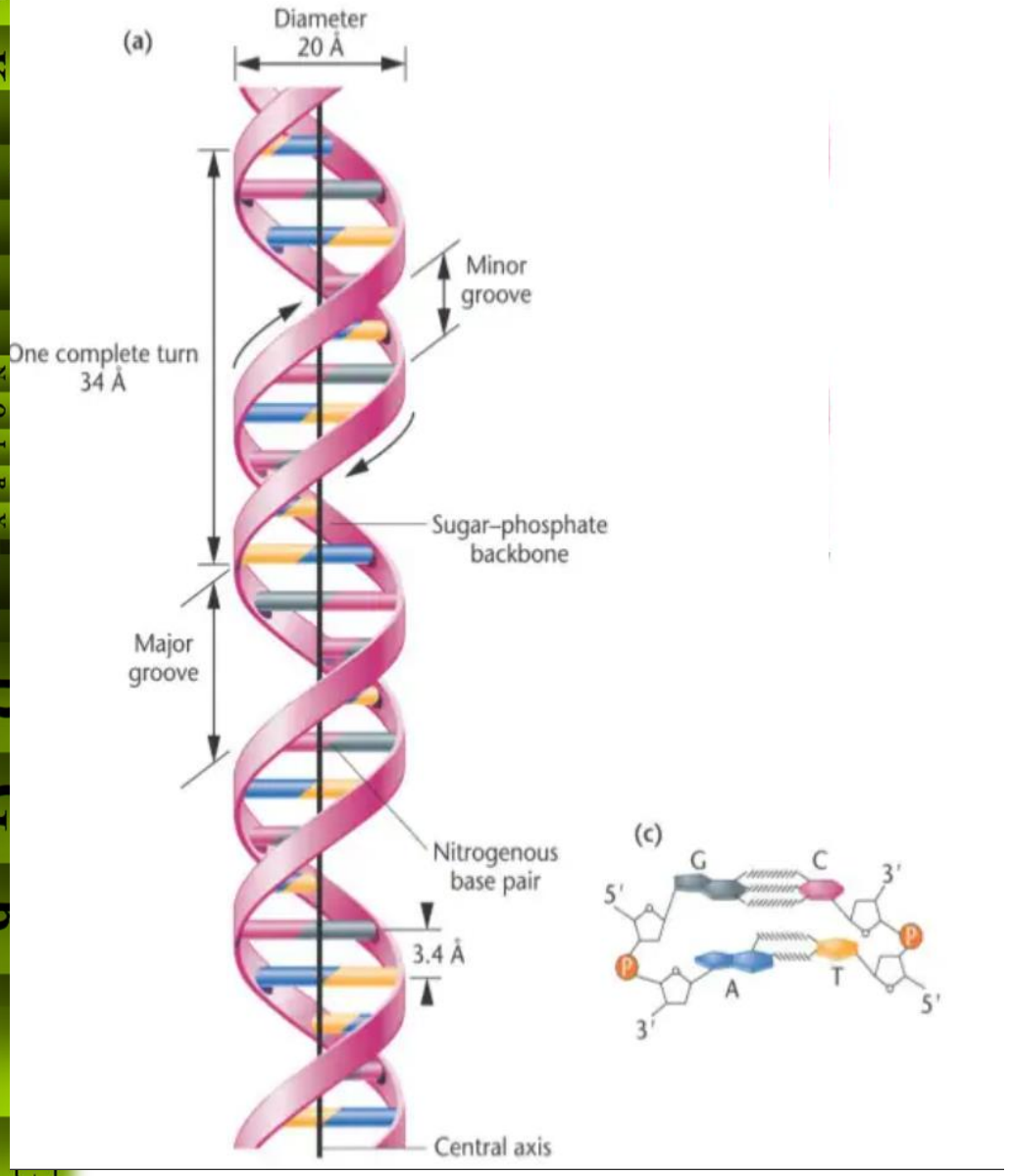
dsDNA structure is antiparallel



Studies shows that parallel orientation is very unstable!

https://www.ncbi.nlm.nih.gov/pmc/articles/PMC101824/

dsDNA structure



Each dsDNA consist of:

1) A phosphate backbone
2) Purines binding with pyrimidine via hydrogen bonds (A-T and G-C)

B-form of DNA (most common):

- Helix has right-handed sense.
- Wide major groove easily accessible to proteins & narrow minor groove.
- Base pairs nearly perpendicular to helix axis
- One spiral is 3.4nm or 34Å.
- Distance between two H-bonds is 0.34nm or 3.4Å.

https://www.pearson.com/channels/ko-kr/genetics/study-guides/dna-structure-and-analysis-foundations-of-5

Different forms of DNA:

B-form

A-form

Z-form

Factors involved for different DNA conformations:

There are at least three factors on which the DNA conformation depends:

1) Ionic or hydration environment

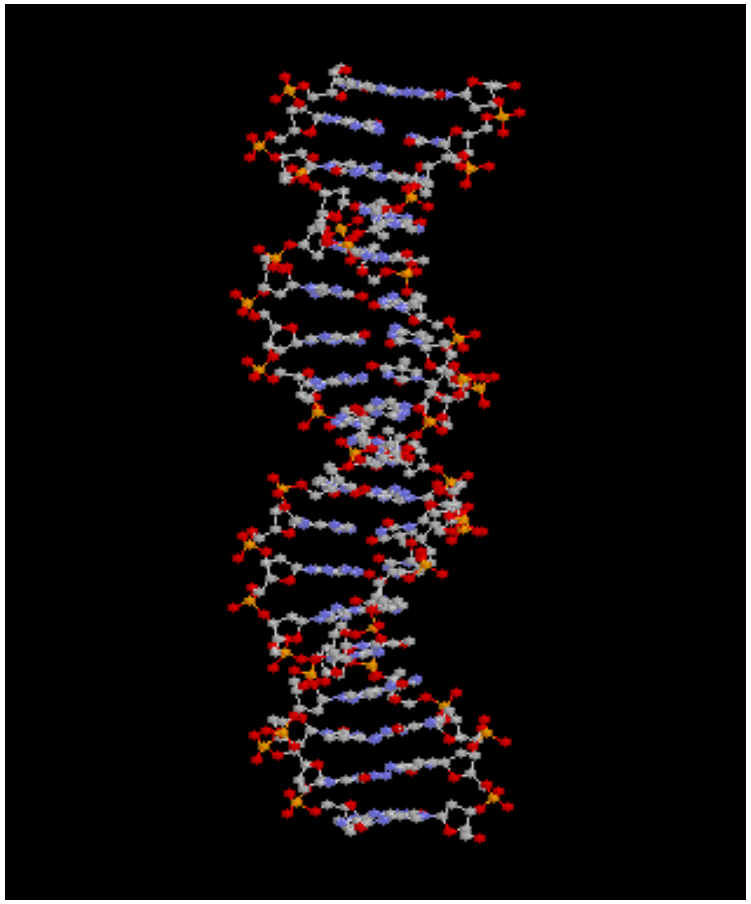
2) DNA sequences.

3) Presence of specific proteins

In a living cell, DNA can occur in the form of A-DNA or B-DNA, with a few regions capable of forming Z-DNA. There can be section of DNA that can locally be in A-form, while the rest of the DNA remains in B-form.

B-DNA

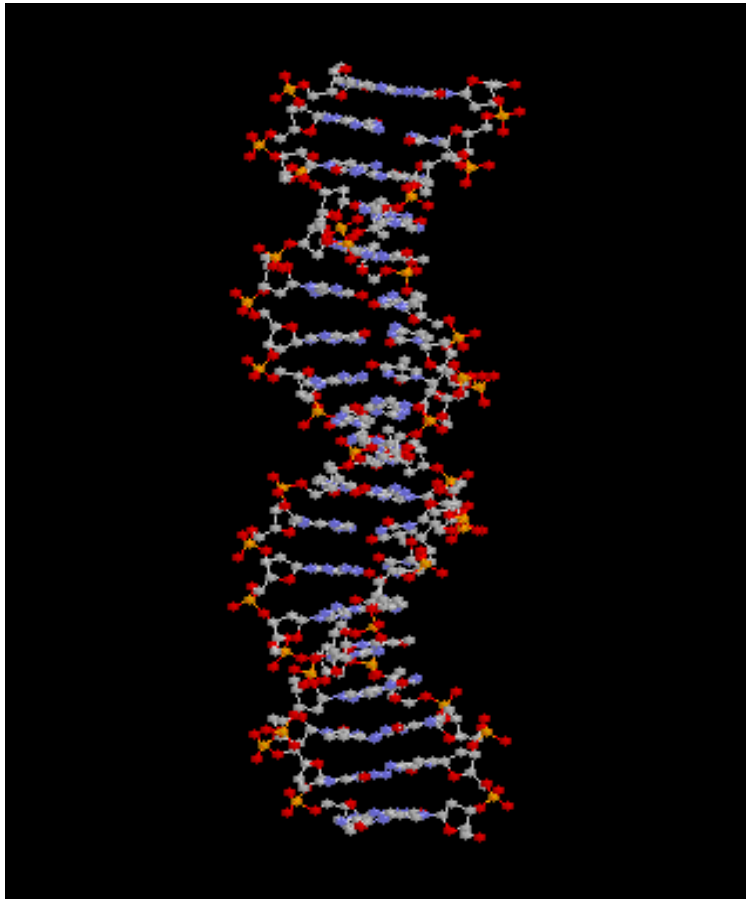
- Most common DNA conformation in vivo.
- Favoured conformation at high water concentrations.
- Also known as Watson & Crick model of DNA.
- First identified in fibre at 92% relative humidity.



The B- DNA structure

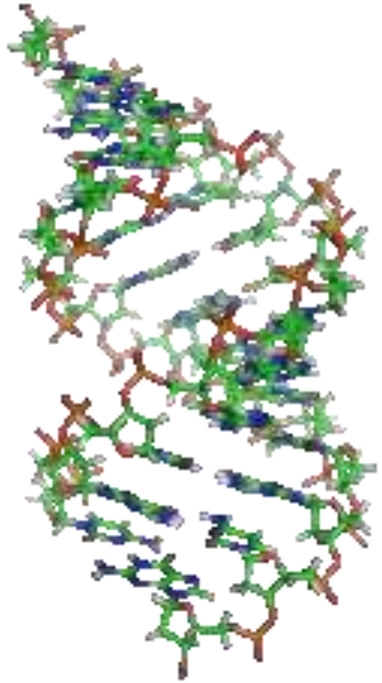
B-DNA structure

- Helix has right handed sense.
- Wide major groove easily accessible to proteins & Narrow minor groove.
- Base pairs nearly perpendicular to helix axis
- One spiral is 3.4nm or 34Å.
- Distance between two H-bonds is 0.34nm or 3.4Å.



A-DNA

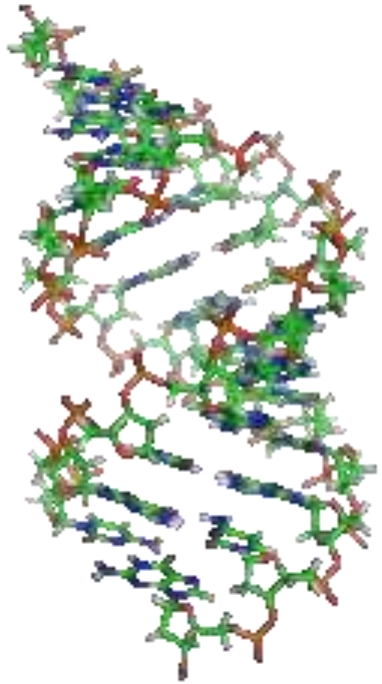
- **A-DNA** is one of the many possible double helical structures of DNA.
- Helix has right-handed sense, shorter more compact helical structure.
- It occurs only in dehydrated samples of DNA, such as those used in crystallographic experiments.



A-DNA structure

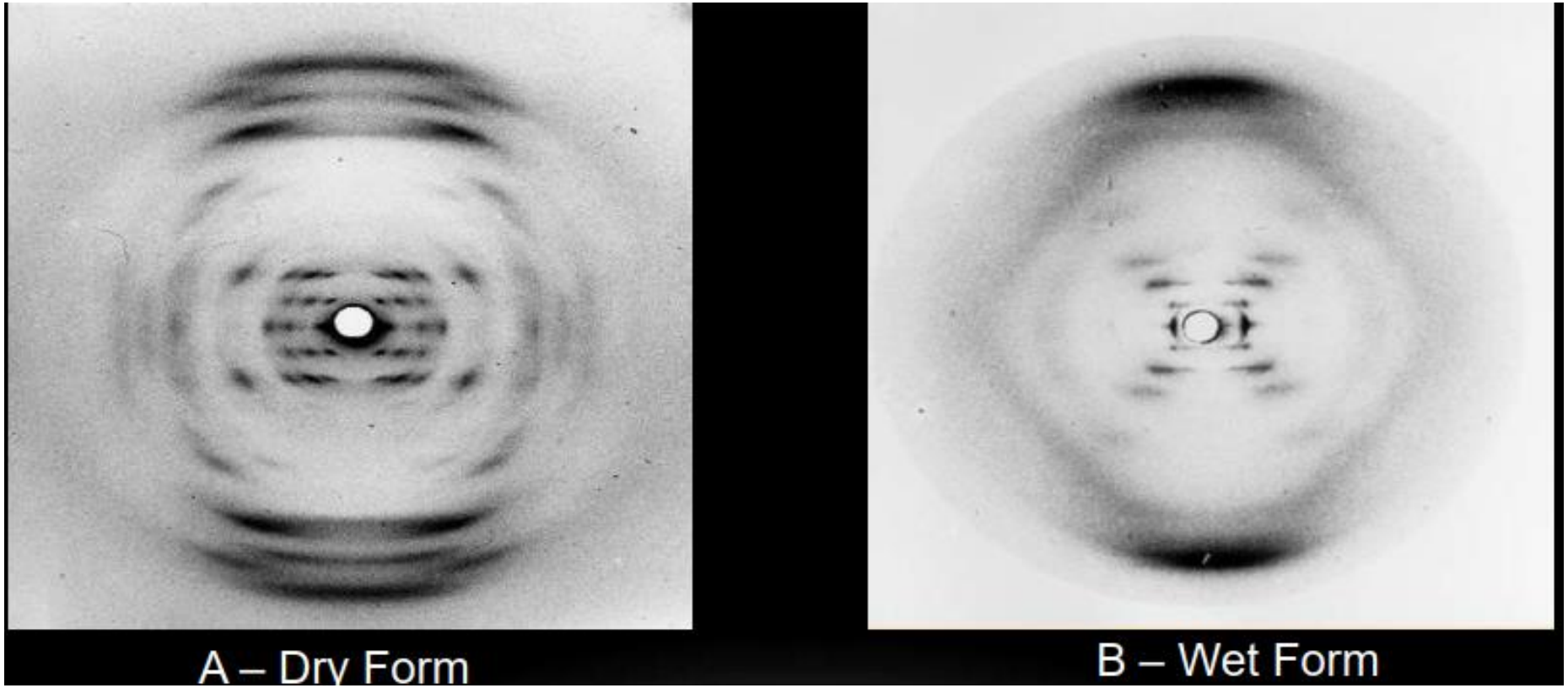
A-DNA structure

- A-DNA is fairly similar to B-DNA.
- Slight increase in the number of bp/ rotation (resulting in a tighter rotation angle), and smaller rise/turn.
- deep major groove and a shallow minor groove.
- Favoured conformation at low water concentrations.
- In a solution with higher salt concentrations or with alcohol added, the DNA structure may change to an **A form**, which is still right-handed, but every 2.3 nm makes a turn and there are 11 base pairs per turn.



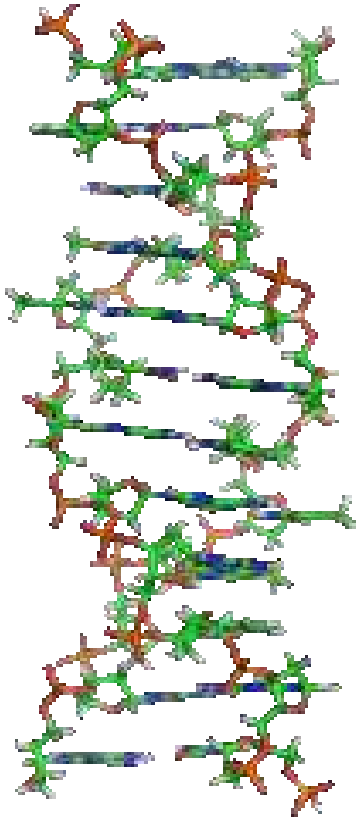
A-DNA structure

Franklin's work



Z-DNA

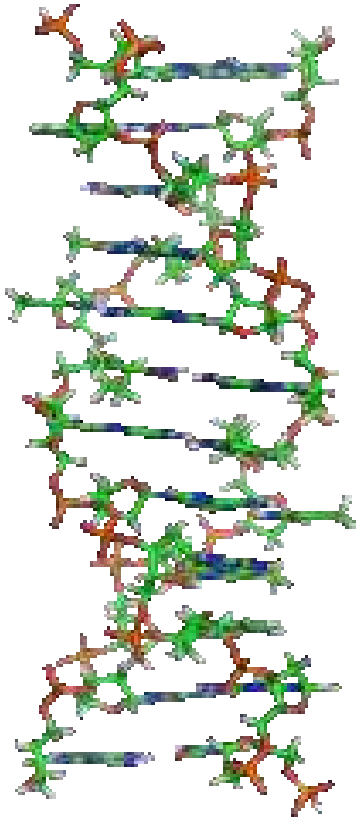
- **Z-DNA** is one of the many possible double helical structures of DNA.
- Helix has left-handed sense.
- Can be formed in vivo, given proper sequence and super helical tension.



Z-DNA structure

Z-DNA structure

- It is a transient form of DNA.
- Narrower, more elongated helix than A or B.
- One turn spans 4.6 nm, comprising 12 base pairs.
- The DNA molecule with alternating G-C sequences in alcohol or high salt solution tends to have such structure.



Z-DNA structure

DNA-forms comparison

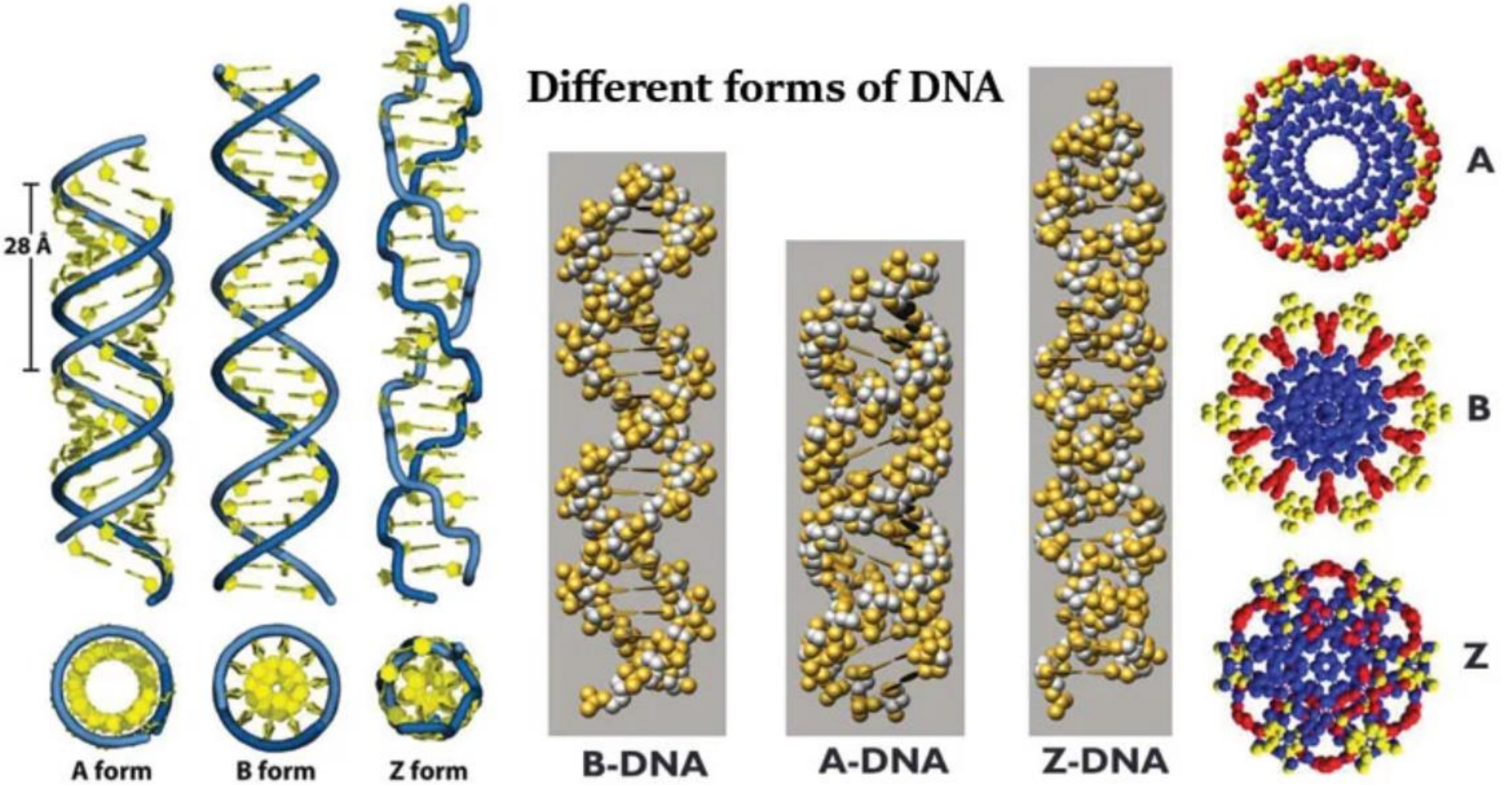


Image source: https://microbenotes.com/different-forms-of-dna-b-form-a-form-z-form/